



LAB REPORT

Course Code : ICE 3146 **Course Title :** Microprocessor and Interfacing Laboratory

Lab Task No. : 08

Lab Task Topic: Introduction to Arduino UNO and Interfacing with a Buzzer.

No.	Evaluation Criteria with Marks	Put Tick (✓) Mark					Marks Obtained	Remarks
		Excellent	Good	Fair	Poor	Fail		
REPORT WRITING								
1	Theory & Objectives (2.5)							
2	Materials & Methods (5)							
3	Data Analysis & Discussion/Diagram (10)							
4	Precautions and Conclusion (2.5)							
5	Time Management (5)							
	TOTAL							
LAB PERFORMANCE								
1	Preparation for Lab (5)							
2	Experiment and Engagement (15)							
3	Communication (5)							
	TOTAL							
15 th July, 2026 Date of Experiment		25 th July, 2026 Date of Submission					Teacher's Signature	

Semester : Summer **Year :** 2026 **Level-Term :** L2-T2 **Section :** A2

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Title: Introduction to Arduino UNO and Interfacing with a Buzzer.

Abstract:

This experiment was performed using the Arduino UNO board and Arduino IDE software to understand the basic concept of microcontroller programming and hardware interfacing with a buzzer. The main objective of this experiment was to become familiar with the Arduino UNO, learn how to write and upload a simple program through Arduino IDE, and control a buzzer through a digital output pin. In this experiment, the buzzer was connected to digital pin 8 of the Arduino UNO, and a simple program was uploaded so that the buzzer remained ON for 3 seconds and OFF for 2 seconds repeatedly. As a result, the buzzer produced a repeating alert sound similar to a ship siren pattern. This experiment helped to understand digital output control, timing using delay, program uploading, and simple interfacing between Arduino UNO and an external output device.

Introduction (Theory):

Arduino UNO is a widely used microcontroller development board based on the ATmega328P microcontroller. It is commonly used in embedded systems and electronics laboratories because it is easy to program and can interface with a large number of external devices. The Arduino UNO contains digital input/output pins, analog input pins, power pins, and a USB interface through which a program can be uploaded from a computer by using Arduino IDE software.

Interfacing means connecting an external hardware component with the microcontroller in such a way that the controller can control or communicate with that device. A buzzer is an electronic output device that converts electrical signals into sound. It is widely used in alarm systems, warning indicators, timers, and notification circuits. When the buzzer is connected to a digital output pin of the Arduino UNO, the microcontroller can control it by sending HIGH and LOW logic signals. A HIGH signal turns the buzzer ON, and a LOW signal turns it OFF.

In this experiment, a buzzer was connected to the Arduino UNO and controlled by a simple program written in Arduino IDE. The pinMode() instruction was used to configure the buzzer pin as an output pin, the digitalWrite() instruction was used to turn the buzzer ON and OFF, and the delay() function was used to control the time duration of each state. Because the buzzer was kept ON for 3000 milliseconds and OFF for 2000 milliseconds, the output produced a repeating sound pattern similar to a ship siren. This experiment demonstrated the basic working principle of Arduino programming, digital output control, timing control, and simple hardware interfacing.

Material:

Hardware	Software
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1. Arduino UNO board & USB cable 2. Buzzer 3. Breadboard 4. Jumper wires 5. Personal Computer (PC) / Laptop	1. Arduino IDE
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Method:

Circuit Connection:

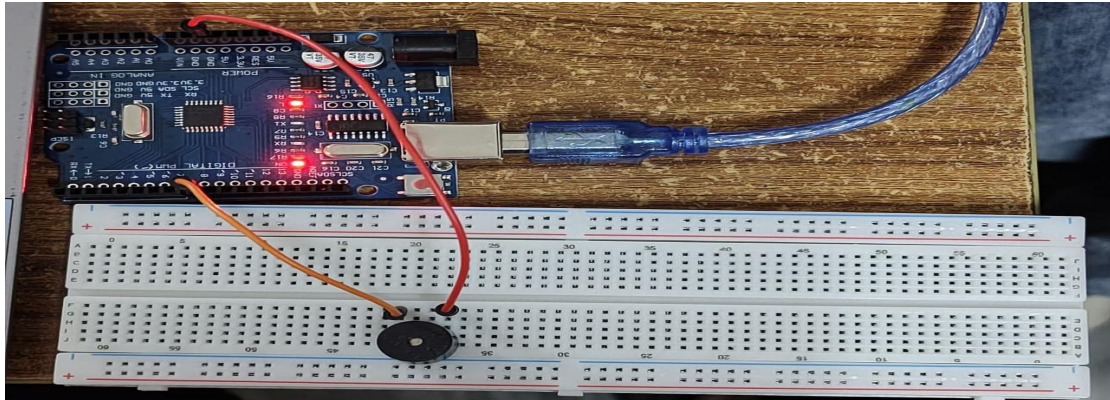


Fig 1: Circuit connection of Arduino UNO and Buzzer.

Code:

```
const int buzzerPin = 8;

void setup()
{
  pinMode(buzzerPin, OUTPUT);
}

void loop()
{
  digitalWrite(buzzerPin, HIGH);
  delay(1000);

  digitalWrite(buzzerPin, LOW);
  delay(1000);
}
```

Execution:

1. The buzzer was placed on the breadboard.
2. The positive terminal of the buzzer was connected to Arduino digital pin 8.
3. The negative terminal of the buzzer was connected to Arduino GND.
4. The Arduino UNO was connected to the computer using a USB cable.
5. The program was written in the Arduino IDE.
6. Arduino UNO and the correct COM port were selected.

7. The program was verified and uploaded.
8. The buzzer turned ON for one second when Pin 8 was HIGH.
9. The buzzer turned OFF for one second when Pin 8 was LOW.
10. The ON and OFF sequence was repeated continuously.

Result:

Output Sequence:

Step	Pin 8 State	Buzzer Condition	Time Duration
1	HIGH	Buzzer ON	1 Second
2	LOW	Buzzer OFF	1 Second
3	HIGH	Buzzer ON	1 Second
4	LOW	Buzzer OFF	1 Second

Discussion:

In this experiment, the basic features of the Arduino UNO and its digital output operation were studied. A buzzer was connected to digital pin 8 and controlled using the digitalWrite() function. When the pin was set to HIGH, the buzzer produced sound, and when it was set to LOW, the buzzer stopped.

The delay(1000) function maintained each condition for one second. Since the instructions were placed inside the loop() function, the ON and OFF sequence was repeated continuously. The experiment demonstrated digital pin configuration, buzzer interfacing, program uploading, and time-controlled output using the Arduino UNO.